Weekly Homework 1

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Problem 1. Compute $\pi/4$ through the partial sum

$$S_{2n} = \sum_{k=1}^{2n} \frac{(-1)^{k-1}}{2k-1}$$

of the Gregory–Leibniz series and estimate the truncation error. Can you propose a simple correction to S_{2n} based on your error estimate?

(optional) Estimate the truncation error again for the corrected scheme.

Solution. Let $r_{2n} = \sum_{k=2n+1}^{+\infty} \frac{(-1)^{k-1}}{2k-1}$ be the truncation error of the partial sum S_{2n}. Notice that

$$\frac{1}{2k-1} = \int_0^1 x^{2k-2}\,\mathrm{d}x.$$

Therefore,

$$\begin{aligned}
r_{2n} &= \sum_{k=2n+1}^{+\infty} (-1)^{k-1} \int_0^1 x^{2k-2}\,\mathrm{d}x \\
&= \int_0^1 \sum_{k=2n+1}^{\infty} (-1)^{k-1} x^{2k-2}\,\mathrm{d}x \\
&= \int_0^1 \sum_{k=2n+1}^{\infty} (-x^2)^{k-1}\,\mathrm{d}x \\
&= \int_0^1 \frac{(-x^2)^{2n}}{1+x^2}\,\mathrm{d}x \\
&= \int_0^1 \frac{x^{4n}}{1+x^2}\,\mathrm{d}x.
\end{aligned}$$

When n is large, the integral above is dominated by the contribution from the left neighborhood of 1. Therefore, let $x = 1 - \frac{t}{4n}$. Then,

$$x^{4n} = \left(1 - \frac{t}{4n}\right)^{4n} \approx e^{-t}, \quad 1 + x^2 \approx 2.$$

Hence,

$$r_{2n} = \frac{1}{8n} + O\left(\frac{1}{n^2}\right), \quad \text{as } n \to \infty.$$

Therefore, a simple correction to S_{2n} is

$$S_{2n} + \frac{1}{8n}.$$

For the corrected scheme, the truncation error is $O(1/n^2)$ asymptotically. $\square$

Problem 2. In principle, the sine function can be evaluated through the Taylor series expansion

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

Let us consider two computational schemes to evaluate the sine function.

(a) Directly truncate the Taylor series. Make sure that the truncation error is less than the rounding error bound for any input x.

(b) First shift x to the interval $(-\pi/2, \pi/2]$, and then apply scheme (a).

Scheme (b) is in general more accurate. Can you provide a theoretical analysis?

Sample at least 1000 points in $[-50, 50]$ (e.g., using the MATLAB/Octave statement `linspace(-50, 50, 1000)`) and plot the errors (e.g., using the MATLAB/Octave function `semilogy`). Can you explain the results?

Proof. By Taylor expansion, we have

$$\sin x = \sum_{k=0}^{n} (-1)^k \frac{x^{2k+1}}{(2k+1)!} + \frac{(-1)^{2n+3} \cos \xi}{(2n+3)!} x^{2n+3}.$$

Shifting x to the interval $(-\pi/2, \pi/2)$ reduces the absolute value of x^{2n+3} and thus reduces the truncation error in general. In particular, when x is large, the truncation error of scheme (a) can be very large, while the truncation error of scheme (b) is still small.

You can see the error plot below. The error of scheme (a) grows rapidly as $|x|$ increases, while the error of scheme (b) remains small for all x.

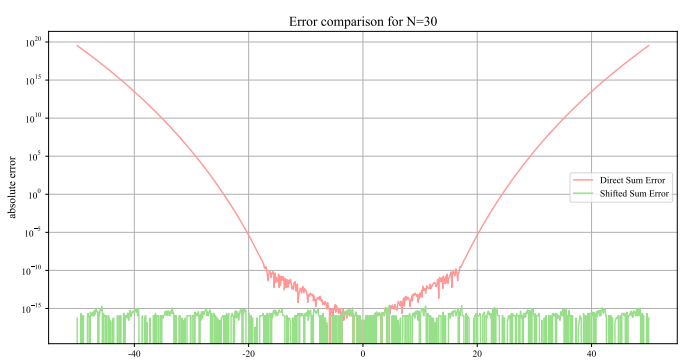

Figure 1: Error plot for scheme (a) and scheme (b).

$\square$

Problem 3. The Lambert W-function, $y = W(x)$, is the inverse function of $x = y\exp(y)$ for $y \in [-1, +\infty)$. Make a plot of the Lambert W-function, with at least 100 equally spaced sampling points over x. Verify your plot using the graph of $x = y\exp(y)$ with equally spaced sampling points over y.

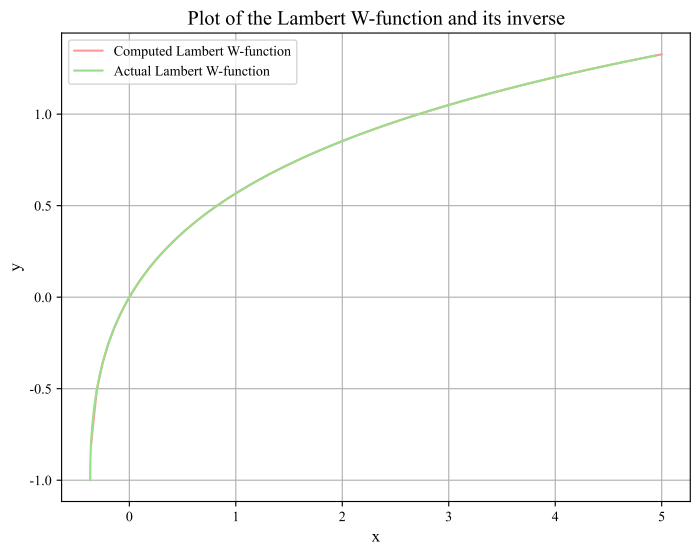

Figure 2: Plot of the computed and actual Lambert W-function

Solution. We use Newton's method to compute the solution of the equation $x = y\exp(y)$ for each x. The actual Lambert W-function is plotted using the function $x = y\exp(y)$ by plotting y against x. The plot of the computed Lambert W-function and the actual Lambert W-function are shown in the figure above. We can see that the computed Lambert W-function matches well with the actual Lambert W-function, which verifies our plot. $\square$

Problem 4. Using Newton's method to find the root of $\arctan x = 0$ is an overkill, since the unique solution, $x_* = 0$ is trivial. However, this is a good example to see that the convergence of Newton's method relies on the initial guess. The set of real initial guesses such that Newton's method converges to x_* is the form $(-\alpha, \alpha)$, where $\alpha > 0$. Try to calculate α with at least 10 significant decimal digits. What happens if α is used as the initial guess?

Solution. We will show that the sequence $\{x_k\}_{k\in\mathbb{N}}$ generated by Newton's method will oscillate between α and $-\alpha$ when we choose α as our initial guess. First, notice that the sequence $\{x_k\}_{k\in\mathbb{N}}$ is sign-alternating, i.e., $x_k x_{k+1} < 0$. This is because the update formula of Newton's method is given by

$$x_{k+1} = x_k - (1 + x_k^2)\arctan x_k.$$

Let's consider the function $f(x) = x - (1 + x^2)\arctan x$. The derivative of this function is $f'(x) = -2x\arctan x \leq 0$, which means that f is monotonically decreasing. Therefore, when

$x_k > 0$, we have $x_{k+1} = f(x_k) < 0$; when $x_k < 0$, we have $x_{k+1} = f(x_k) > 0$. Hence, the sequence $\{x_k\}_{k\in\mathbb{N}}$ is sign-alternating.

From the result above, we can conclude that

$$|x_{k+1}| = (1 + |x_k|^2) \arctan |x_k| - |x_k|$$

since $f(x)$ is an odd function. Let $y_k = |x_k|$. Now we show that the sequence $\{y_k\}$ converges to 0 iff the sequence is monotonically decreasing. Suppose that there exists some $k \in \mathbb{N}$ such that $y_{k+1} > y_k$. Then, we have

$$(1 + y_k^2) \arctan y_k > 2y_k.$$

Let $\alpha > 0$ be the root of $g(y) = (1 + y^2) \arctan y - 2y$. Then $g(y) > 0$ for all $y > \alpha$ and $g(y) < 0$ for all $y \in (0, \alpha)$. Now that $g(y_k) > 0$, we have $y_k > \alpha$; then $y_{k+1} > y_k > \alpha$, which indicates that $y_{k+2} > y_{k+1} > \alpha$, and so on. Therefore, the sequence $\{y_k\}$ is monotonically increasing and hence $\{x_k\}$ can never converge since it is sign-alternating.

On the other hand, if the sequence $\{y_k\}$ is monotonically decreasing, since the sequence is bounded below by 0, it must converge to some limit $l \in [0, \alpha)$. Taking the limit on both sides of the update formula, we have

$$l = (1 + l^2) \arctan l - l \iff g(l) = 0 \implies l = 0.$$

Therefore, the sequence $\{x_k\}$ converges to 0 if the sequence $\{y_k\}$ is monotonically decreasing.

Now we can calculate α by finding the root of $g(y)$ using Newton's method. The computed solution of α is The root alpha ¿ 0 of the function g(y) is approximately: 1.3917452002707351, with at least 10 significant decimal digits.

When we choose α as our initial guess solution of $\arctan x$, we have $g(\alpha) = 0$. Therefore, the sequence $\{y_k\}$ generated by Newton's method will be constant, i.e., $y_k = \alpha$ for all $k \in \mathbb{N}$. Hence, the sequence $\{x_k\}$ will oscillate between α and $-\alpha$. $\square$

Problem 5. Write a program to plot the Newton fractal for the cubic equation $x^3 - 1 = 0$ over the region $\{z \in \mathbb{C} : -2 \leq \mathrm{Re}(z) \leq 2 \text{ and } -2 \leq \mathrm{Im}(z) \leq 2\}$.

(optional) You are encouraged to try other polynomial equations and other regions in the complex plane.

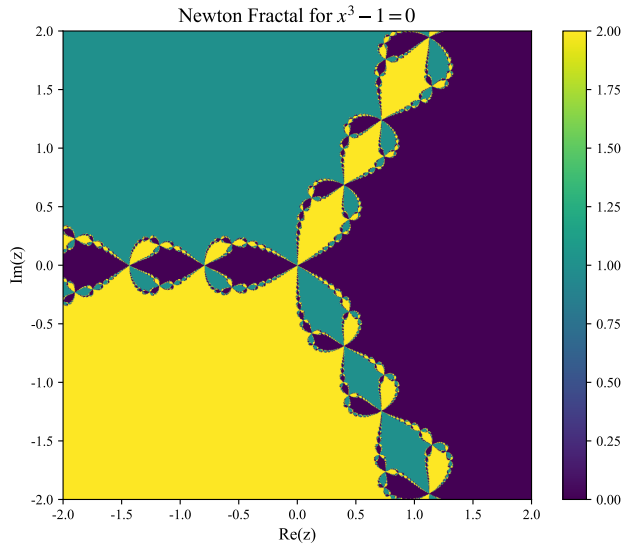

Figure 3: Newton fractal for the cubic equation $x^3 - 1 = 0$.

Problem 6. (optional) Solve the following problem from pintia.cn and explain your algorithm.

Solution. Notice that $\psi(1) = \sum_{k=1}^{+\infty} \frac{1}{k(k+1)} = 1$. Now we can subtract $\psi(1)$ from $\psi(x)$ to get

$$\psi(x) - \psi(1) = \sum_{k=1}^{+\infty} \frac{1}{k(k+x)} - \frac{1}{k(k+1)} = \sum_{k=1}^{\infty} \frac{1-x}{k(k+1)(k+x)}.$$

We denote the right-hand side of the equation above by $S_1(x)$. Similarly, we subtract $S_1(1)$ from $S_1(x)$ to get $S_2(x)$; then we subtract $S_2(1)$ from $S_2(x)$ to get $S_3(x)$; and so on. After m iterations, we have

$$\psi(x) = \sum_{i=0}^{m-1} C_i S_i(1) + (1-x)^m S_m(x),$$

where $C_i = S_i(1)$ and $S_m(x) = \sum_{k=1}^{\infty} \frac{1}{k(k+1)^m(k+x)}$.

The first term on the right-hand side can be computed easily; as for the second term, consider

$$\sum_{k=n}^{\infty} \frac{1}{k^r} < \int_{n-1}^{\infty} \frac{1}{x^r}\,\mathrm{d}x = \frac{(n-1)^{1-r}}{r-1}, \quad r > 1.$$

Thus the truncation error of the second term is less than $\frac{1}{(m+1)(n-1)^{m-1}}$. Therefore, let $m = 10$, we get $n = 12$. Therefore, we only need to compute 12 terms in the series $S_{10}(x)$ to get an approximation of $\psi(x)$ with an error less than 0.5×10^{-12}. $\square$

Problem 7. (optional) Design an algorithm to compute the reciprocal of a nonzero real number (i.e., $\alpha \mapsto \alpha^{-1}$) without any division operation.

Solution. α^{-1} can be viewed as the solution of the equation $f(x) = \frac{1}{x} - \alpha = 0$, which can be solved by Newton's method. The update formula of Newton's method is given by

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} = 2x_k - \alpha x_k^2.$$

Let $x_* = \alpha^{-1}$ and $e_n = x_n - x_*$. Then,

$$e_{n+1} = 2x_n - \alpha x_n^2 - x_* = 2e_n - \alpha x_n^2 - \alpha x_*^2 = -\alpha e_n^2.$$

To make the error e_n converge to 0, we need $|\alpha e_n^2| < |e_n|$ for all n. One possible condition is $|e_0| < \frac{1}{|\alpha|}$.

For the simplicity of the notation, let's consider the case $\alpha > 0$. Otherwise, we can just add a negative sign in the final result.

We can represent the original number α in the form of $\alpha = \beta \cdot 2^e$, where $\beta \in (0.5, 1]$. Now we only need to compute β^{-1} since 2^{-e} is a power of 2 and can be computed simply by bit manipulation. Since $\beta^{-1} \in [1, 2)$, we can choose $x_0 = 1$ as our initial guess, which satisfies the condition $e_0 < \frac{1}{\beta}$.

Then we can apply the update formula iteratively:

$$x_{k+1} = 2x_k - \beta x_k^2, \quad k = 0, 1, 2, \ldots$$

Finally, we can compute α^{-1} by multiplying x_k with 2^{-e}. $\square$